↳ Negtively charged
↳ Anode
↳ Oxidation
↳ loss of electron

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Electro chemistry

Electro chemical cell :-

The device which converts chemical energy into electrical energy or vice versa

Types

1. Electrolytic cell

=) The device which converts electrical energy into chemical energy. (when electricity is supplied chemical reaction takes place)

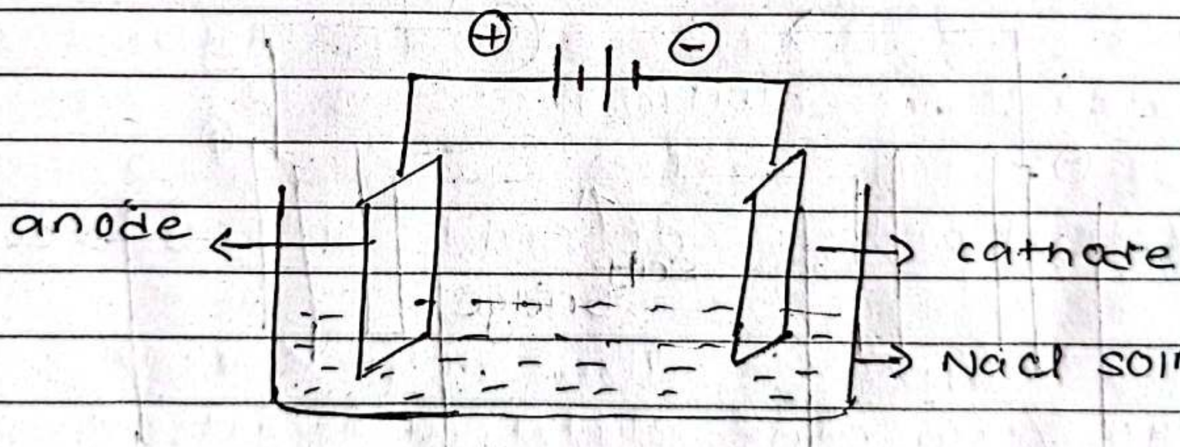

Fig :- Electrolytic cell.

$NaCl \rightarrow Na^+ + Cl^-$

At cathode :- $Na^+ + 2e^- \longrightarrow 2Na$

At anode :- $2Cl^- - 2e^- \longrightarrow Cl_2$

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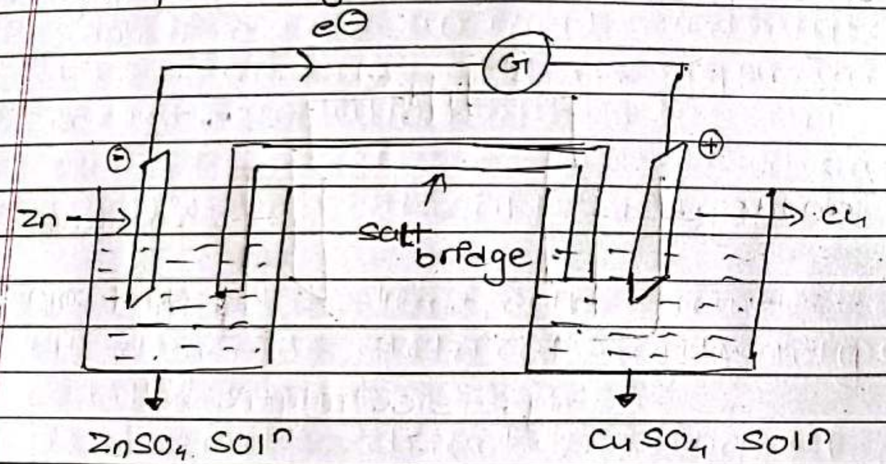
2 Galvanic cell

=) The cell which converts chemical energy into eletrical energy galvanic cell. [Electricity is produced by chemical reaction]

Example of Galvonic cell

- Daniel cell (Zn-Cu) cell is common example of galvonic cell



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In Zn-Cu cell there are two half cells

i) Oxidation half cell.

=) This half consists of zinc electrode dipped in $ZnSO_4$ electrolytic solution. As zinc has lowe reduction potential value (-0.76)v. It acts as anode zinc atoms loes to electrons to electrode and pass into soln as Zn^{++} ions. Thus zinc electrode becomes negtively charged

ii) Reduction half cell

=) This compartment consists of Cu electrode dipped in $CuSO_4$ soln. It acts as cathode as it has high reduction potential value (+0.34) volt. Here Cu^{++} ions gain two electrons and convert to Cu atoms.

~~This~~ These two half cells are externally connected by connecting wire with galvonometer and internally connected by salt bridge. Electrons flow from anode to cathode and electricity is generated.

At anode (Oxdn)

$Zn - 2e^- \rightarrow Zn^{2+}$

At cathode (Redn)

$Cu^{2+} + 2e^- \rightarrow Cu$

Net cell rxn : $Zn + Cu^{++} \rightarrow Zn^{++} + Cu$

Salt bridge

⇒ It is invented V shaped glass tube which consist of inert electrolyte like KCl, KNO_3 etc in agar agar paste or solution.

Application of salt bridge

1. It completes the cell circuit.
2. It maintains eletrical neutrality of cell.
3. It prevents direct mixing of electrolytic solution.
4. It minimus liquid junction potential.

cell notation.

⇒ The symbolic representation of galvanic cell is called cell Notation.

eg Anode / Anode soln // cathode soln / cathode

phase boundry   salt bridge

$Zn / Zn^{2+} // Cu^{2+} / Cu$

Q. Write cell notation.

i. $Ni + Cu^{2+} \rightarrow Ni^{+2} + Cu$

ii. $Mg + 2Ag^{+} \rightarrow Mg^{2+} + 2Ag$

⇒ Solution

i $Ni / Ni^{++} // Cu^{2+} / Cu$

ii $Mg / Mg^{2+} // 2Ag^{+} / 2Ag$

Q. Write rxns at anode and cathode and net cell

a. $Cu / Cu^{++} // Ag^{+} / Ag$

At Anode $Cu - 2e^{\ominus} \rightarrow Cu^{++}$

At cathode $2Ag^{+} + 2e^{\ominus} \rightarrow 2Ag$

Net rxn : $Cu + 2Ag^{+} \rightarrow Cu^{++} + 2Ag$

b. $Al / Al^{+++} // Cu^{++} / Cu$

At Anode:- $2Al - 6e^{\ominus} \rightarrow 2Al^{++}$

At cathode:- $3Cu^{++} + 6e^{\ominus} \rightarrow 3Cu$

Net rxn:- $2Al + 3Cu^{++} \rightarrow 2Al^{+++} + 3Cu$

Single Electrode Potential

=) When Metal electrode is dipped in it's electrolytic solution. Three cases may occur:-

i. Metal ions may collide to surface of metal electrode but no change take place.

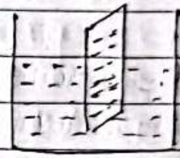
ii. Metal ions may collide to electrode surface and gain e⁻s [electrons]

$M^{n+} + ne^{\ominus} \rightarrow M$

iii. Metal atoms may loose e⊖s and form metal ions.

$M - ne^{-} \rightarrow M^{n+}$

The electrode potential developed at electrode / electrolyte interface due to seperation of charge [Formation of electrical double layer] is called single electrode potential or electrode potential.



Single electrode potential cannot be measured practically. To measure the electrode potential of cell it must be connected with reference electrode.

- Oxidation potential :- The tendency of metal to get oxidize is called oxidation potential.

- Reduction potential :- The tendency of metal ions to get reduced is called reduced potential.

=) oxidation potential and reduction potential have same value but opposite sign.

E°_{oxd} ~~eg~~:- $Zn / Zn^{++} = +0.76\ V$

E°_{redn} $Zn / Zn^{++} = -0.76\ V$

standard electrode potential

=) The electrode potential of a cell measured at standard conditions (Temp = 25°C, pressure = 1 atm and molar concentration of electrolytic solution). standard electrode potential of an electrode is measured by coupling it with standard hydrogen electrode.

standard Hydrogen electrode [SHE] or [NHE]

=) It is a primary refrence electrode. It consist of glass, ~~rod~~ wood with platinum foil dipped in 1M HCl solution and H_2 gas is passed at 1 ATM pressure. It's electrode potential is 0 volt.

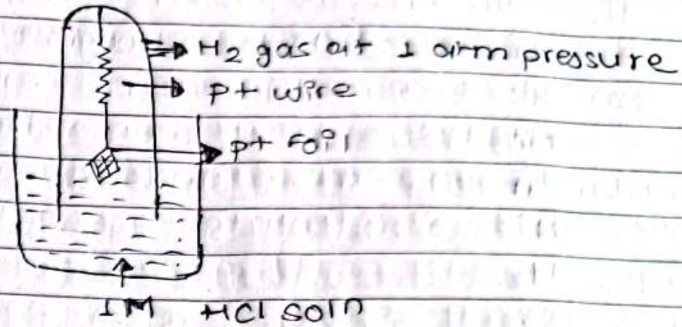


Fig:- SHE / NHE

- For Zn - SHE cell [Zn = -0.76 volt]

  $Zn / Zn^{++} // H^+ / H_2, Pt$ [Zn - anode]

- For Cu - SHE cell [$E_{Cu/Cu^{++}} = +0.34V$]

  $Pt, H_2 / H^+ // Cu / Cu^{++}$ [SHE - anode]

Electrochemical series. [ECS]

=) The series of metal electrodes arrange vertically in order of their increasing standard reduction potential values is called electrochemical series.

[reducing tendency] AAfu oxidize hune arulai reduce garne

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Li -3.05V strongest reducing agent
K
Mg
Zn
Ni
H_2
Cu
Ag
F_2

Application of electro chemi

1. To predice anode and cathode of galvonic cell
2. The electrode with reduction potential value acts as cathode while electrode with reduction potential value acts as anode.
3. To find em.f of cell. The difference electrode potential of two half cells is called electro motive force or shell potential

Formula $\boxed{E^\circ cell = E^\circ cathode - E^\circ anode}$

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$E^\circ Zn/Zn^{++} = -0.76V$
$E^\circ Cu/Cu^{++} = +0.34V$
$= 0.34 - (-0.76)$
$= 1.1$ volt

4. To find the whether the cell is spontoneus or not.

[If em.f of cell is positive, cell is spontoneus whereas If em.f of cell is negative cell is Non-spontoneus]

5. To calculate oxidising and reducing strength. The metals above a hydrogen is ECS (i.e low redⁿ potential) are reducing agents and metals below hydrogen in ECS (i.e high redⁿ potential are oxidising agent.

[Li is the strongest reducing agent and Florine is the strogent oxidising agent]

6. To predict metal displacement reactions.
=) The metals with low reduction potential value can display metals with high redⁿ potential value from their salt solution

$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
$ZnSO_4 + Cu \rightarrow \times$ [NO rxn]
[because Cu has high reduction potential]

~~Numerical~~ NERNST Equation

⇒ It is a mathematical expression which relates emf at non-standard condition to emf at standard condition, tempr and electrolyte concentration.

$$\boxed{E_{cell} = E^\circ_{cell} - \frac{2.303\,RT}{nF}\log Q}$$ ← use when tempr is other than 25°C

When tempr = 298 K / 25°C

$$\boxed{E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log Q}$$

E_{cell} = Emf of cell of non-standard condition

E°_{cell} = EmF of cell of standard condition

n = no of $e^\ominus$s involved

F = 96500 coulomb

R = Universal gas constant (8.314)

Q = equilibrium constant = $\frac{[C]^c [D]^d}{[A]^a [B]^b}$

Numerical

Q. Calculate em.f of cell at 25°C when concentration of $ZnSO_4$ and $CuSO_4$ are 0.001M and 0.1M respectively

$E^\circ_{Zn/Zn^{++}} = -0.76V$

$E^\circ_{Cu/Cu^{++}} = +0.34V$

Solution,

$Zn / Cu^{++} \rightarrow Zn^{++} + Cu$

$Zn / Zn^{++} (0.001M) // Cu^{++} (0.1M) / Cu$

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$= 1.1V$

$E_{cell} = E^\circ_{cell} - 0.0591 \log \frac{[Zn^{++}][Cu]}{[Zn][Cu^{++}]}$

$= 1.1V - \frac{0.0591}{2} \log \frac{[0.001][1]}{[1][0.1V]}$

$= 1.1591V$ #

[anode - oxidation
loss of electron]

Solid meta Zn = 1M

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From the following data

$E^\circ\ Fe^{+++}/Fe^{++} = 0.77V$

$E^\circ\ Hg^{++}/Hg = 0.79V$

$[Fe^{++}] = 0.2M$ $[Fe^{+++}] = 1M$ and

$[Hg^{++}] = 0.4M$

write i) electrode rxn?

ii) net cell rxn?

iii) Emf of cell

iv) cell notation

v) spontaneity of cell.

Solution,

At anode $2Fe^{++} - e^- \rightarrow Fe^{+++}$

At cathode $Hg^{++} + 2e^- \rightarrow Hg$

~~net cell rxn~~ :-

$2Fe^{++} - 2e^- \rightarrow 2Fe^{+++}$ (multiply by 2)

net cell rxn

$2Fe^{++} + Hg^{++} \longrightarrow 2Fe^{+++} + Hg$

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emf of cell = cathode - anode

$(E^\circ) = 0.79 - 0.77$

$= 0.02V$

cell notation =

$2Fe^{++} / 2Fe^{+++} // Hg^{++} / Hg$

$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2} \log \frac{[Fe^{+++}]^2 [Hg]}{[Fe^{++}]^2 \cdot [Hg^{++}]}$

$= 0.02 - \frac{0.0591}{2} \log \frac{[1]^2 \cdot \frac{1}{2}}{(0.2)^2 \cdot 0.4}$

$= -0.033V$

It is non spontoneous.

what is concentration of Ni^{++} ion in a cell at 25°C if emf of cell is 0.601 V

Imp
asked
2times

$Ni / Ni^{++} (x=?) // Cu^{++} (0.75M) / Cu$

$E^\circ Ni/Ni^{++} = -0.25V$

$E^\circ Cu/Cu^{++} = +0.34V$

solution,

E.m.f of cell (E°) = cathode - anode
= $0.34 - (-0.25)$
= 0.59 V

At anode

$Ni - 2e^- \rightarrow Ni^{++}$

At cathode

$Cu^{++} + 2e^- \rightarrow Cu$

$$E = E^\circ - \frac{0.0591}{2} \log \frac{[1][Ni^{++}]}{[0.75][1]}$$

$$0.601 = +0.59 - 0.02955 \log \frac{[Ni^{++}]}{[0.75]}$$

$$-0.3722 = \log \frac{[Ni^{++}]}{0.75}$$

$Ni^{++} = 0.3183M$

[... series read] common metal.

calculate cell emf at 35°C

$E^\circ\ Fe|Fe^{++} = 0.44V \rightarrow -0.44V$

$E^\circ\ Ag|Ag^+ = -0.80V$

(oxidation - reduction) = $+0.80V$

E.m.f of cell E° = cathode - anode
= $0.80 - (-0.44)$
= 1.24 V

At anode

$Fe - 2e^- \rightarrow Fe^{++}$

At cathode

$(Ag^+ + e^- \rightarrow Ag) \times 2$

$2Ag^+ + 2e^- \rightarrow 2Ag$

$Fe + 2Ag^+ \rightarrow Fe^{++} + 2Ag$

net cell rxn

$$E_{cell} = E^\circ_{cell} - \frac{2.303\ RT}{nF} \log Q$$

$$= 1.24 - \frac{2.303 \times 8.314 \times 308}{2 \times 96500} \log Q$$

$$= 1.24 - \frac{2.303 \times 8.314 \times 308}{2 \times 96500} \log$$

$= 1.24V.$ #

Q. A radio uses 3.12V for its operation. From given material how can you operate radio.

Given,

a) Zn rod

b) Ag rod

c) 0.2M $ZnSO_4$ soln

d) 1M $AgNO_3$ soln

$E^\circ_{Zn/Zn^{++}} = +0.76V = -0.76V$

$E^\circ_{Ag^+/Ag} = +0.80V$

Anode rxn :- $Zn - 2e^\ominus \rightarrow Zn^{++}$

cathode rxn : $[Ag^+ + e^\ominus \rightarrow Ag] \times 2$

$2Ag^+ + 2e^\ominus \rightarrow 2Ag$

Net cell rxn : $Zn + 2Ag^+ \rightarrow Zn^{++} + 2Ag$

$E^\circ \Rightarrow E^\circ_{cathode} - E^\circ_{anode}$

$\Rightarrow 0.8 - (-0.76)$

$\Rightarrow 1.56V$

~~Again~~

Again,

$$E = 1.56 - \frac{0.0591}{2} \log \frac{[0.2][1]^2}{[1][1]^2}$$

$= 1.56 - 0.014$

$= 1.5V$

From 2 cell of Zn and Ag we can operate the radio.

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Electro chemical series

F_2	+ 2.87
H_2O_2	+ 1.77
Au^+	+ 1.68
Cl_2	+ 1.36
O_2	+ 1.23
Br_2	+ 1.09
Ag^-	+ 0.80
Fe^{3+}	+ 0.77
I_2	+ 05.4
O_2	+ 0.40
Cu^{2+}	+ 0.34
S	+ 0.14
$2H^-$	0.00
Pb^{2+}	− 0.13
Sn^{2+}	− 0.14
Ni^{2+}	− 0.23
Co^{2+}	− 0.28
Fe^{2+}	− 0.44
Zn^{2+}	− 0.76
H_2O	− 0.83
Mn^{2+}	− 1.03
Al^{3+}	− 1.67
Mg^{2+}	− 2.34
Na^+	− 2.71
Ca^{2+}	− 2.87
K^+	− 2.93
Li^+	− 3.025

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Q. A Zn rod is placed in 0.1 M solution of $ZnSO_4$ at 25°C. calculate potential of zinc electrode.

$E^\circ_{Zn|Zn^{++}} = -0.76\,V$

solution,

For single electrode rxn

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{1}{Zn^{++}}$$

$$= -0.76 - \frac{0.0591}{2} \log \frac{1}{(0.1)}$$

$$= -0.78\,V$$ #

Q. A Zn rod is placed in 0.1M solution of $ZnSO_4$ at 25°C and salt is dissociated to 95% extent. Calculate the electrode potential of zinc electrode

$E^\circ_{Zn/Zn^{++}} = -0.76\,V$

$$\text{Conc}^n \text{ of } Zn^{++} = \frac{95}{100} \times 0.1$$

$$= 0.095$$

For single electrode rx^n

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{1}{Zn^{++}}$$

$$= -0.76 - \frac{0.0591}{2} \log \frac{1}{0.095}$$

$$= -0.790\ V$$ #

CORROSION

=) The slow and continuous deterioration or destruction of metal surface due to its interaction with environmental component (water, gases, chemicals etc) is called corrosion.

Rusting of iron is common example of corrosion. The broun amorphous scale or powder formed on the surface of metal due to it's interaction with environmental components (water, gases, chemical etc) is called rust. (Hydrated ferric oxide) $Fe_2O_3 \cdot xH_2O$

electrochemical theory of corrosion.

=) At non-uniform surface of iron a part of it's surface with low red^n potential acts as anode and the part with high red^n potential acts as cathode. The reactive gases like CO_2, SO_2 etc along with moisture (water) act as electrolyte. In this way an electro chemical cell is set up at surface of iron.

At anode, $Fe - 2e^- \rightarrow Fe^{++}$

At cathode, $H_2O + \frac{1}{2} O_2 + 2e^- \rightarrow 2OH^-$

$$2Fe^{++} + 2OH^- \rightarrow 2Fe(OH)_2$$

$$2Fe(OH)_2 \xrightarrow{\text{oxidation}} 2Fe(OH)_3 \xrightarrow{\text{decomposition}} Fe_2O_3 \cdot 3H_2O$$

(rust)

At anode Fe loses $2e^-$s to form Fe^{++} ions and at cathode hydroxyl ions are formed. They combine to form ferrous hydroxide which further oxidises to form Ferric hydroxide. It decomposes to $Fe_2O_3 \cdot 3H_2O$ (rust)

BATTERY :-

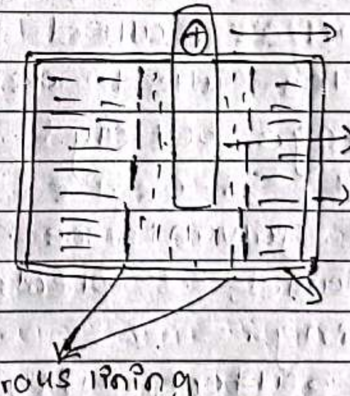
The two or more galvanic cells connected in series is called battery. They are also called commercial cells. They must be cheap and portable.

Types of commercial cells / battery.

1. Primary cell :-

The cell in which spontaneous chemical rxn takes place in one direction and cannot be recharged is called primary cell.

Dry cell is a common example of primary cells. Primary cells are used in toys, calculators, watches, torch light etc.



Dry cell consists of zinc container which acts as anode of graphite rod surrounded by $(MnO_2 + C)$ powder which acts as cathode separated by porous lining with $NH_4Cl + ZnCl_2$ paste which act as electrolyte.

At anode,

$$Zn - 2e \rightarrow Zn^{++}$$

At cathode,

$$2MnO_2 + 2NH_4^+ + 2e^- \rightarrow Mn_2O_3 + 2NH_3 + H_2O$$

Secondary cell:-

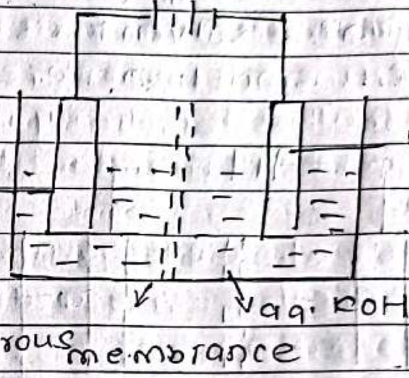
i) The cell which can be ~~recha~~ recharged again and again is called secondary cell. It is also called as rechargable battery or storage battery.

It works as galvanic cell during discharging and as electrolytic cell during charging.

eg:- Ni-cd battery, lithium-ion battery and sodium ion battery etc.

1. Ni-Cd cell:-

This secondary cell consists of Cd (cadmium) as anode and NiO_2 as cathode and aq. KOH as electrolyte.



It is widely used in electrical appliances and engines of aeroplanes and helicopters.

Working:

At anode,

$$Cd^{++} + 2OH^- \rightarrow Cd(OH)_2 + 2e^-$$

At cathode,

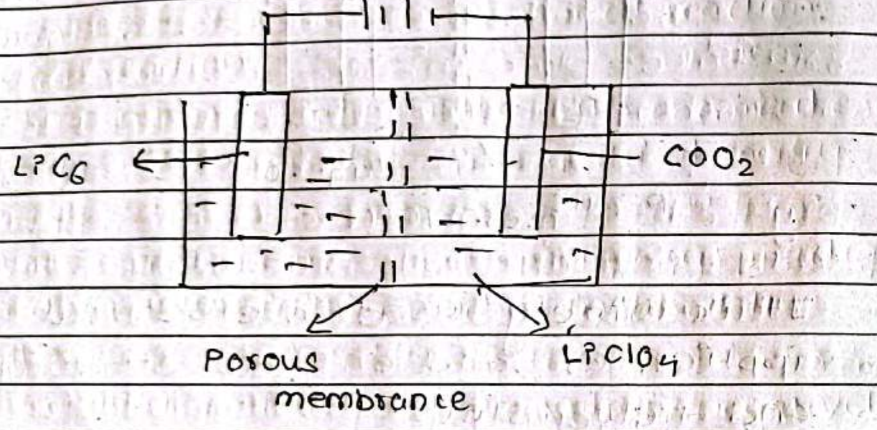
$$NiO_2 + 2H_2O + 2e^- \rightarrow Ni(OH)_2 + 2OH^-$$

Net cell rxn:-

$$Cd^{++} + NiO_2 + 2H_2O \rightarrow Cd(OH)_2 + Ni(OH)_2$$

2. Li-ion battery.

⇒ This secondary cell consists of LiC_6 (lithium intercalated carbon electrode) as anode and CoO_2 (cobalt oxide) as cathode & $LiClO_4$ (lithium perchlorate) in organic solvent as electrolyte. It is used in smart phone, laptop etc.



working

At anode:-

$LiC_6 \rightarrow Li^+ + C_6 + e^-$

At cathode:-

$CoO_2 + Li^+ + e^- \longrightarrow LiCoO_2$

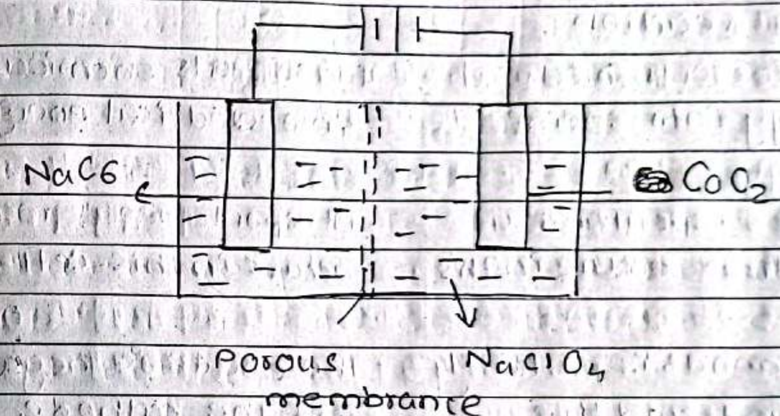
Net cell rxn:-

$LiC_6 + CoO_2 \longrightarrow C_6 + LiCoO_2$

It is very portable, easy to charge, and also known future generating battery.

3. Na - ion battery

This secondary cell consists of NaC_6 (sodium intercalated carbon electrode) as anode & CoO_2 (cobalt oxide) as cathode and $NaClO_4$ (sodium perchlorate) in organic solvent as electrolyte. It is widely used in electrical vehicles and solar equipments etc.



Working:-

At anode

$NaC_6 \longrightarrow Na^+ + C_6 + e^-$

At cathode.

$CoO_2 + Na^+ + e^- \longrightarrow NaCoO_2$

Net cell rxn:-

ASSIGNMENT NO: 1

Write difference between electrolytic and galvanic cell. Calculate emf of cell at 20°C when $E^\circ Fe + Fe^{++} = +0.44V$, $E^\circ cu/cu^{+} + 0.34V$

Electrolytic cell	Galvanic cell
* An electrolytic cell is a cell that uses an electric current for the progression of a chemical reaction	A galvanic cell is an electro-chemical cell that can produce electricity with the help of a chemical reaction.
* It converts electrical energy into chemical energy.	It converts chemical energy into electrical energy.
* A non-spontaneous reaction occurs in this cell.	A spontaneous reaction occurs in this cell.
* The anode is positively charged, and the cathode is negatively charged.	The anode is negatively charged and the cathode is positively charged.

Numerical

Given,

$E^\circ Fe/Fe^{++} = +0.44V \Rightarrow -0.44V$ (acc to ECS)

$E^\circ cu/cu^{++} = -0.34V \Rightarrow +0.34V$ (acc to ECS)

EmF = ?

At anode : $Fe - 2e \rightarrow Fe^{++}$

At cathode $cu^{++} + 2e \rightarrow Cu$

we know,

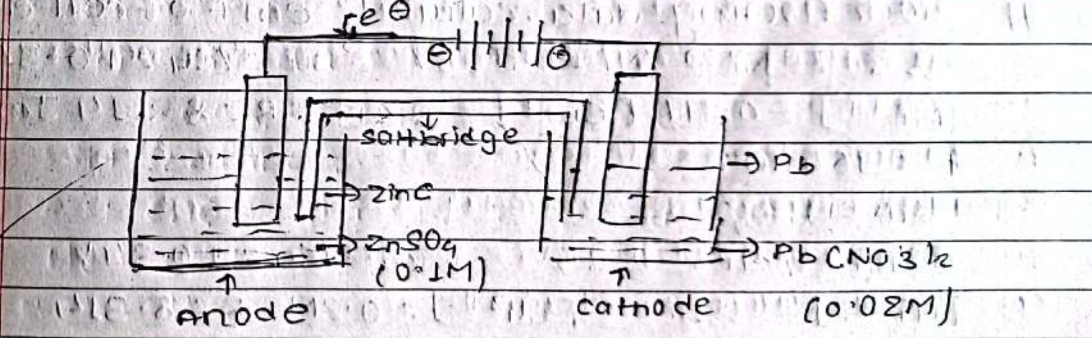
$$E^\circ = E^\circ cathode - E^\circ anode$$
$$= 0.34 - (-0.44)$$
$$= 0.78V$$

Now,

$$EmF = E^\circ - \frac{2.303RT}{nF} \log Q$$

$$= 0.78 - \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \log \frac{[1][1]}{[1][1]} = 0.78V \#$$

Q.2 A galvanic cell consists of two electrode, zinc and lead, zinc electrode is immersed in 0.1 M zinc ion soln and lead electrode in 0.02M lead ion soln. Calculate emf of cell at 298K write eqns of electrode process and represent cell schematically. Given that $E^\circ (Zn^{++}/Zn) = -0.76V$. $E^\circ (Pb^{++}/Pb) = -0.13V$.



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Given:
$E^\circ(Zn^{++}/Zn) = -0.76V$ $E^\circ(Pb^{++}/Pb) = -0.13V$
(0.1M) (0.02M)

At anode:
$Zn - 2e^- \rightarrow Zn^{++}$

At cathode
$Pb^{++} + 2e^- \rightarrow Pb$

Net rxn: $Zn + Pb^{++} \rightarrow Zn^{++} + Pb$

Net cell rxn: $Zn/Zn^{++} // Pb^{++}/Pb$
(0.1M) (0.02M)

$E^\circ cell = E^\circ cathode - E^\circ anode$
$= -0.13V - (-0.76V)$
$= 0.63V$

Now,
$$EmF = E^\circ - \frac{2.303RT}{nF} \log Q$$
$$= 0.63 - \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \log \frac{[Zn^{++}][Pb]}{[Zn][Pb^{++}]}$$

$$= 0.63 - \frac{2.303 \times 8.314 \times 298}{2 \times 96500} \log \frac{[0.1]}{[0.02]} = 0.6V \#$$

From flowing data. calculate emf of cell at 25°C $E^\circ Ni^{2+}/Ni = -0.25V$ $E^\circ Cu^{2+}/Cu = +0.34V$ $[Ni^{++}] = 0.1M$ $[Cu^{2+}] = 0.2M$. $R = 8.314 \ Jmol^{-1}K^{-1}$ $F = 96500C$

Given,
$E^\circ Ni^{2+}/Ni = -0.25V$ $E^\circ Cu^{2+}/Cu = +0.34V$
$[Ni^{++}] = 0.1M$ $[Cu^{2+}] = 0.2M$ $R = 8.314$

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Rxn at cathode : $Cu^{++} + 2e \rightarrow Cu$
Rxn at anode : $Ni - 2e \rightarrow Ni^{++}$
Net cell rxn: $Ni + Cu^{++} \rightarrow Ni^{++} + Cu$

$Emf = E^\circ cathode - E^\circ anode$
$= 0.34 - (-0.25) = 0.59V$

Now,
$$Emf = E^\circ - \frac{2.303RT}{nF} \log Q$$
$$= 0.59 - \frac{2.303 \times 8.314 \times 303}{2 \times 96500} \log \frac{[0.1]}{[0.2]}$$
$= 0.599V \#$

write advantage and disadvantage of dry cell, Ni-Cd battery, lithium ion battery, Na-ion battery and Daniel cell.

Advantages of dry cell:

1. Dry cell is small in size and light in weight too due to which it can be transported from place to place. ~~which it can be~~ There is no risk of leakage of chemicals in a dry cell.
2.
3. Dry cell are Inexpensive
4. electrolyte used in dry cell is not harmful for the environment.

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Disadvantage of dry cell.

1. It cannot be recharged once it loses its electrical power.
2. voltage drops due to build up of reaction products around the electrode as when current is drawn rapidly from it.
3. The self life of dry cell is short (6-8 months)

Advantage of Ni-cd battery.

1. It delivers high current output
2. It is relatively tolerant of overcharging.
3. Ni-Cd batteries perform well in a wide range of temperature, making them suitable for both hot and cold environments.
4. Ni-Cd batteries can be charged quickly, making them suitable for applications that require rapid replenishment of energy.

Advantage of lithium ~~battery~~ ion battery

- It has longest life.
- More usable capacity
- constant Power
- Temperature Tolerant
- Charging - Fast and safe

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Disadvantages of Ni-cd battery

1. It has high self-discharge rate (1% per day)
2. It's cost is high and due to use of cadmium it is harmful.
3. It is three to five times more expensive than lead - acid.

Disadvantage of lithium ion battery

1. It involves risk of bursting.
2. It is costly. compared to other batteries.
3. complete discharge damage the battery.
4. Extremely sensitive to high temperatures (degrades very quickly, if exposed to heat)
5. very short lifespan (2 to 3 years from the date of manufacturing. even if not in use)
6. Not available in standard cells sizes like others.

Advantage of Na-ion battery

1. It is Rechargable of energy storage.
2. It is Easier to recycle.
3. It has low market prices.
4. It has capable of working at room tempr, Good efficiency.

Disadvantage of Na-ion battery.

1. Large ionic size of Na+ which require more power to keep energy flowing.
2. lower operation voltage.
3. lower energy density.
4. Need high temperature for optimal work.

Advantage of Daniel cell.

1. simple design and construction
2. Reliable voltage output
3. long-lasting cell life
4. Inexpensive materials used
5. Minimal maintenance required.

Disadvantage of deniell cell

1. limited energy capacity
2. Not rechargable
3. Bulky and heavy.
4. slow energy delivery
5. corrosion over time.

5. Write difference between.

a. Primary cell and secondary cell.

	Primary cell	secondary cell
1.	Primary cells are batteries that cannot be recharged or reused	secondary cells are batteries that can be recharged and reused
2.	can be used only once	canebe used more than once
3.	used in portable devices as they produce current immediately.	Needs to be charged before use and are in automobiles
4.	Have lower self discharge rates and can be used for long term storage of power	Have a higher self discharge rate compared to primary cells.

b. Electrochemical and chemical corrosion

Electrochemical	chemical corrosion
1. It occurs in the presence of moisture or electrolyte	It occurs in dry condition.
2. It is due to formation of large number of anodic and cathodic areas.	It is due to direct chemical attack of the metal by the environment.
3. Heterogeneous (bimetallic) surface alone get corroded	even a homogenous metal gets corroded
4. It is a continuous process.	It is a self controlled process.
5. It follows electro chemical reaction	It adopts absorption mechanism.
6. Rusting of iron in ~~moisture~~ moist atmosphere is an example	Formation of mild scale on iron surface is an example.

Q. Write effects and prevention of corrosion.

Ans Effects of corrosion are:-

1. loss of efficiency.
2. contamination of product.
3. Damage of metallic equipments.
4. Inability to use metallic materials.
5. Accidents due to mechanical lose of metallic bridges, cars, aircrafts etc.
6. causes pollution due to escaping products from corrosion.
7. Depletion of natural resource (Metals)

Prevention of corrosion.

1. By plating the metal with a thin layer of less easily oxidized metal like tin
2. By coating the metal with more active metals like zinc i.e Galvanisation steel with zinc.
3. By allowing a protective film of metal oxide to form naturally on the surface
4. By applying a protective coating of paint or enamel on the surface of metal.

Date
Page

Q. What is the concⁿ of Ni^{+2} ion in cell at 27°C, if emf is 0.601 V? (Ni(s) / Ni^{+2} (?) // Cu^{2+} (0.75 M) / Cu(s) standard electrode potential of Nickel and copper are −0.25V and +0.34 V respectively.

Given

emf = 0.601V

Temp^r = 27°C = 305K

$E^\circ_{cathode}$ = +0.34V

E°_{anode} = −0.25 V

Cu^{2+} = 0.75M

Ni^{+2} = ?

At cathode: $Cu^{++} + 2e \rightarrow Cu$

At Anode: $Ni + 2e \rightarrow Ni^{++}$

Net: $Cu^{++} + Ni \rightarrow Cu + Ni^{++}$

We know,

$E^\circ_{mf} = E^\circ_{cathode} - E^\circ_{anode}$

$= 0.34 - (-0.25) = 0.59\ V$

Now,

$Emf = E^\circ - \frac{2.303RT}{nF} \log Q$

$\Rightarrow 0.601 = 0.59 - \frac{2.303 \times 8.314 \times 305}{2 \times 96500} \log \frac{[1][Ni^{++}]}{[0.75]}$

$\Rightarrow 0.011 = -0.03 \log \left[\frac{Ni^{++}}{0.75}\right]$

$\Rightarrow -0.367 = \log \left[\frac{Ni^{++}}{0.75}\right]$

$\Rightarrow 10^{-0.367} = \frac{[Ni^{++}]}{0.75}$

$= 0.4295 \times 0.75 = [Ni^{++}]$

$\therefore [Ni^{++}] = 0.322M$

∴ The concⁿ of Ni^{+2} ion is 0.322 M.

Date
Page

8. Calculate emf of given cell.

$Sn/Sn^{++}(0.2M) // Ag^+ (0.1M) / Ag$, where E° sol Sn^{++} = 0.14V and $Ag / Ag^+ = -0.80V$

Ans Acc to ~~Ecs~~, ECs,

$Ag / Ag^+ = +0.80V$ and $Sn / Sn^{++} = 0.14V$

Now,

Rxn at cathode: $(Ag^+ + e \rightarrow Ag] \times 2$

Rxn at Anode: $Sn - 2e \rightarrow Sn^{++}$

$E^\circ = E^\circ_C - E^\circ_A = -0.8 - (0.14) = +0.94V$

Now

$E_{cell} = E^\circ - \frac{0.0591}{n} \log \frac{[Sn^{++}][Ag]^2}{[Sn][Ag^+]^2}$

$= +0.99 - \frac{0.0591}{2} \log \frac{[0.2][1]^2}{[1][0.1]^2}$

$= +0.94 - 0.038$

$= +0.902\ V$

∴ The emf of given cell is +0.902V

Seen
2080-9-13

substrate = ... molecule]

Organic Chemistry

Nucleophile (nucleus loving species)

=) The species (nucleus loving species) which are $e^{\ominus}$ rich and attack at $e^{\ominus}$ deficient (i.e +ve) centres are nucleophiles. For eg :- OH^-, CN^-, $H_2\ddot{O}$, $\ddot{N}H_3$ etc

Nucleophilic substitution rxns

=) The substitution rxns initiated by nucleophile is called Nucleophilic substitution reaction. There are two types of reaction.

1. S_N1 rxn
2. S_N2 rxn

1. S_N1 reaction.

=) It stands for unimolecular nucleophilic substitution rxn. The reaction whose rate depends only on concentration of substrate (alkyl halide) is called S_N1 rxn

Rate $\propto$ [substrate]

Rate = k [substrate]

ionization ↓ break down into ions] [cation = +ve charge] [carbo cation = +ve charge in carbon]

It follows 1st order kinetics.

For eg :- Hydrolysis of 3° alkyl halide

$$CH_3-\underset{CH_3}{\overset{CH_3}{C}}-Br + \overset{\oplus}{Na}OH^{\ominus} \rightarrow CH_3-\underset{CH_3}{\overset{CH_3}{C}}-OH + NaBr$$

2-bromo-2methyl propane, 2-methyl propan-2-o[l]

(CN) tert-butyl bromide (C.N) tert - butyl alcoh[ol]

Mechanism :-

It involves 2 step:-

1. Formation of carbocation

The slow ionization of substrate forms carbocation.

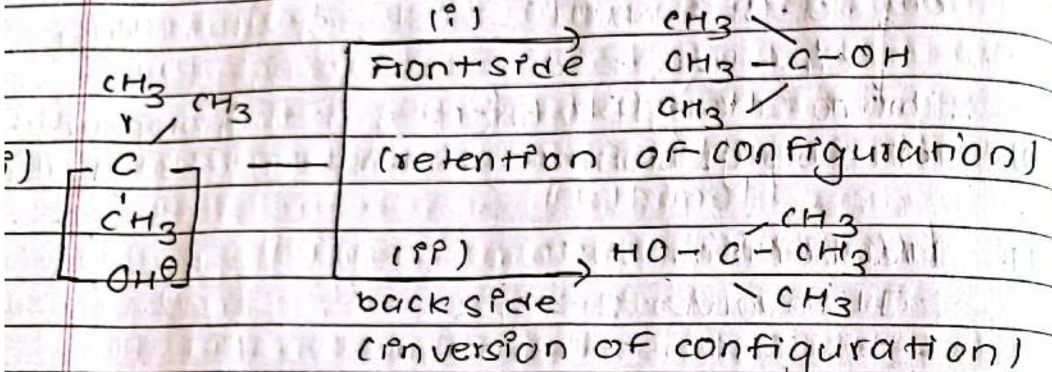
$$\underset{\text{(Substrate)}}{CH_3-\underset{CH_3}{\overset{CH_3}{C}}-Br} \xrightarrow[\text{ionization}]{\text{slow}} \underset{\text{tert-butyl carbocation}}{CH_3-\underset{CH_3}{\overset{CH_3}{C}}^{\oplus}} + Br^- \quad \text{(r.d.s)}$$

only substrate participate in this step and this step is called as rate determining step (r.d.s) describe the configuration

[stereo chemistry]

2. Attack of nucleophile :-

⇒ Nucleophile attacks fast to carbon. cation and forms product (alcohol)



⇒ The Front side attack of nucleophile leads to formation of tert - butyl alcohol with retention of configuration i.e configuration of product is similar to configuration of substrate.

⇒ The backside attack of nucleophile to carbocation leads to formation of tert - butyl alcohol (product) with inversion of configuration (opposite to that of substrate). Inverted product is dominant

$\underbrace{3°>2°}_{SN_1 \text{ more favorable}} > \underbrace{1° \text{ methyl}}_{SN_2 \text{ more favorable}}$ ⇒ 3° gives SN_1 reaction and 1° gives SN_2 reaction.

Reactivity of alkyl halides towards SN_1 rxn : $\underbrace{3°>2°}_{\text{more favorable}} > 1° >$ methyl

2. SN_2 reaction

⇒ It stands for bimolecular nucleophilic substitution rxn. The reaction whose rate depends on both conc. of substrate of nucleophile is called SN_2 rxn.

∴ Rate = k [substrate nucleophile]

It follows second order kinetic

For eg:- Hydrolysis of 1° alkyl halide.

$CH_3Br + Na^+OH^{\ominus} \longrightarrow CH_3OH + NaBr$

methyl bromide — Methanol

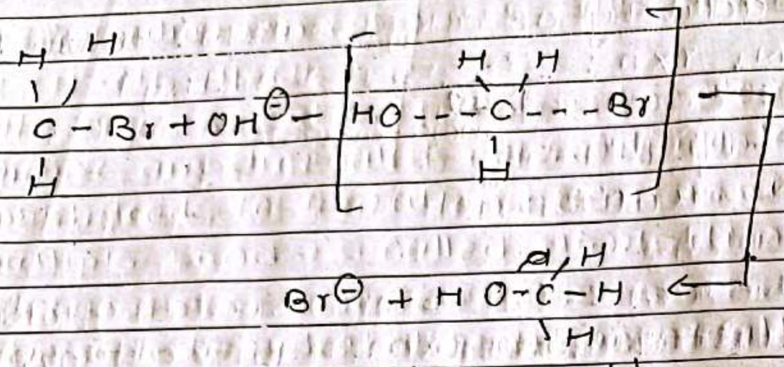
(CN) Bromo methane

Mechanism :-

It involves 1 step through transition state.

[Transition

SN_1 - polar
SN_2 - Non polar



(Methanol)

(inversion of configuration)

As OH^- and Br^- repel each other so nucleophile tends to attack from backside and form T.S which C-OH bond formation and C-Br (breakage) takes place simultaneously. It leads to formation of product with inversion of configuration.

Diff between SN_1 and SN_2

SN_1	SN_2
It stands for unimolecular substitution reaction.	It stands for bimolecular substitution reaction.
two step process	one step process
It follows 1^{st} order kinetics.	It follows 2^{nd} order kinetics
It's rate depend only on conc. of substrate.	It's rate depends on both conc of substrate and nucleophile.
It is more favorable in $3° > 2° > 1° >$ methyl	It is more favorable in $1° > 2° > 3°$
It is typically slower than SN_2	It is faster
favored by weak nucleophiles	favoured by strong nucleophile.
formation of product with inversion of configuration & retention of configuration.	formation of product with inversion of configuration.

Elimination Rxn

The organic reaction in which two atoms or groups are eliminated from adjacent carbon which leads to formation of carbon-carbon multiple bond is called elimination rxn.

For eg:- Dehydrohalogenation of alkyl halide

$$\underset{\text{chloro propane}}{CH_3-CH(H)-CH_2(Cl)} + alc.\ KOH \rightarrow \underset{\text{propene}}{CH_3-CH=CH_2} + KCl + H_2O$$

Two Types:-

1. E_1 rxn:- The elimination rxn in which rate depends only on conc^n of substrate is called E_1 rxn. It is unimolecular ~~ar~~ elimination rxn.

Rate = k[substrate]

It follows 1st order kinetics.

For eg:- E_1 rxn of 2-chloropropane

$$\underset{\text{2-chloropropane}}{CH_3-\overset{Cl}{\overset{|}{C}H}-CH_3} + alc.\ KOH \rightarrow CH_3-CH=CH_2 + KCl + H_2O$$

Mechanism:-

It occures in 2 step:-

1. Formation of carbocation:-
Slow ionization of substrate takes place and carbocation is formed.

$$CH_3-\overset{Cl}{\overset{|}{C}H}-CH_3 \xrightarrow[(r.d.s)]{\text{ionization}} CH_3-\overset{\oplus}{C}H-CH_3$$

2. Abstraction of proton:-
Proton is abstracted by base (attacking reagent) from carbocation which leads to formation of propene.

$$CH_3-\overset{\oplus}{C}H-\overset{H}{\overset{|}{C}H_2} + OH^{\ominus} \longrightarrow \underset{\text{propene}}{CH_3-CH=CH_2} + H_2O$$

Orientation of product in E_1 rxn (saytzeff rule)

=) When E_1 rxn leads to formation of more than one alkene, the product which is highly substituted is the major product. or hydrogen is eliminated from adjacent carbon with least no of H atom.

so major p

less hydroge — more hydrogen

$CH_3-CH_2-CH-CH_3 \xrightarrow[\text{KOH}]{\text{alc}}$

→ $CH_3-CH=CH-CH_3$ But-2-ene (major)

→ $CH_3-CH_2-CH=CH_2$ But-1-ene (minor)

=) When 2-bromobutane under goes E_1 rxn but-2-ene and but-1-ene is formed where but-2-ene is the major product.

E_2 reaction :- (Bimolecular elimⁿ rxn)

=) The elimination rxn in which rate depends ~~of~~ upon concⁿ of substrate and base.

Rate = k[Substrate] [Base]

It follows 2nd order of kineticces.

For eg:- E_2 rxn of bromoethene

$CH_3-CH_2-Br + \text{alc } KOH \longrightarrow CH_3-CH_2 + KBr + H_2O$

Mechanism :-

It occurs in 1 step through transition ~~step~~ state.

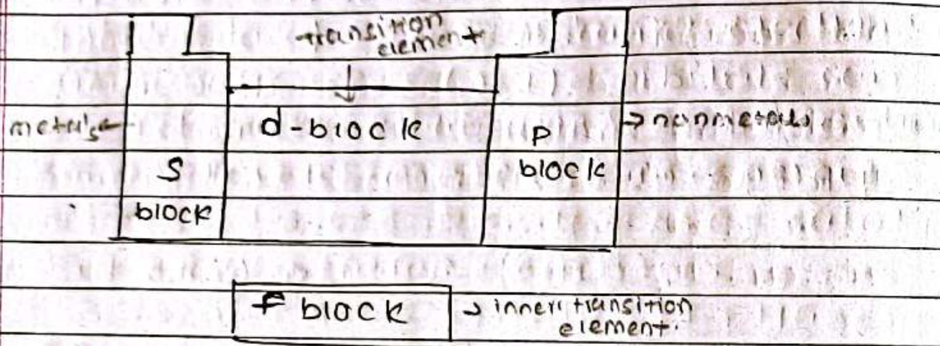
$CH_3-CH_2-Br \xrightarrow{OH^-} \left[\begin{array}{cc} HO-H & Br \\ CH_2 & -CH_2 \end{array} \right]_{T.S} \rightarrow CH_2=CH_2 + H_2O + Br$

=) In T.S the base abstracts hydrogen, C-H and C-Br bond breaks along with formation of double bond between two carbon atoms.

[s block - high tendency to loose election
[p block - high tendency to gain electron

Transition elements

=) The elements which have incompletely filled d-orbitals or f-orbitals (in their metallic or other ondn state) are called transition element. They are d-block (main transition metals) and f-block elements (inner transition elements)



D-block elements lie between s-&p block elements in perodic table. There are 4 series of d-block elements, they are 3d, 4d, 5d and 6d.

characterstics of transition element:

i. Their general outer electronic configuration is $(n-1)d^{1-10}\ ns^{1-2}$

D block element are transition element except zinc

[onidation states => net charge kati x?] means
[dulta metal min vayer alloys form huncha]

ii. They have variable onidation states.

iii. They form coloured compounds.

iv. They form complen compounds.

v. They exhibit catalytic properties.

vi. They form alloys.

1. Electronic configuration

=) The 3d-series consists of elements from atomic no. 21 to 30. The general outer electronic configuration of 3d series electron is $(n-1)d^{1-10}\ ns^{1-2}$ where (n = outer energy level). Among 3d series Cr and Cu have exception at configuration

	3-d elements	electronic configuration
21.	Scandium (Sc)	[Ar] $4s^1\ 3d^1$
22.	Titanium (Ti)	[Ar] $4s^2\ 3d^2$
23.	Vanadium (V)	[Ar] $4s^2\ 3d^3$
24.	Chromium (Cr)	[Ar] $4s^1\ 3d^5$
25.	Maganese (Mn)	[Ar] $4s^2\ 3d^5$
26.	Iron (Fe)	[Ar] $4s^2\ 3d^6$
27.	Cobalt (Co)	[Ar] $4s^2\ 3d^7$
28.	Nikel (Ni)	[Ar] $4s^2\ 3d^8$
29.	copper (Cu)	[Ar] $4s^1\ 3d^{10}$
30.	zinc (Zn)	[Ar] $4s^2\ 3d^{10}$

Note:- [Cr ⇒ $3d^4 4s^2$ but
· $3d^5 4s^1$ half filled d-orbitals are more stable]

same for Cu

Note:- [Cu ⇒ $3d^9 4s^2$ but
$3d^{10} 4s^1$ due to full field
d-orbital are more stable]

Oxidation state :-

:) Transition elements exhibit variable oxidation state (i.e 2 or more than 2 O.S) · It is because electrons from both ns & (n-1)d orbitals can participate in bounding as these two energy levels are close in energy.

3-d element	O.S
Sc	+2, +3
Ti	+2, +3, +4
V	+2, +3, +4, +5
Cr	+1, +2, +3, +4, +5, +6
Mn	+2, +3, +4, +5, +6, +7
Fe	+2, +3, +4, +5, ~~+6~~ +6
Co	+2, +3, +4, +5
Ni	+2, +3, +4
Cu	+ 1, +2
Zn	+ 2

+2 O.S is common oxidation state for transition elements. In 3d series Mn has highest +7 O.S. Among all transition element · Ru & Os have highest +8 O.S.

Ru - Ruthenium.
Os - Osmium

Magnetic property :-

:) Transition elements are mostly paramagnetic in nature. Paramagnetic substance are those substances which are attracted in magnetic field. Diamagnetics substance are those substance which are repelled in magnetic field.

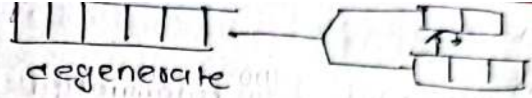
Those metal - ions which have d^0 or d_{10} configuration are diamagnetic in nature. Higher is number of unpaired $e^{\ominus}$s, stronger is the paramagnetic nature.

It is measured in terms of magnetic moment

$$\mu = \sqrt{n(n+2)}\ B.M$$

n = no. of unpaired $e^{\ominus}$s
μ = magnetic moment
B.M = Bohr magneton unit.

To be a transition element must be partially
electron
↓
Tara zinc ma fully filled xa



degenerate

catalytic property:-

⇒ Transition element exhibit catalytic properties. It is because they can show variable oxidation states and they have vacant d-orbitals which provide large surface area for adsorption.

Various transition elements are used as catalyst in chemical rxns.

For eg:-

1. V_2O_5 (vanadium pentoxide) is used as catalyst in contact process.
2. Ni, Pt, Pd etc are used as catalyst in hydrogenation reactions.
3. Fe is used as catalyst in Habers process.

complex compounds formation.

Transition metals can form complex compounds where metal ions is linked with no of negative ions or neutral molecules. This negative ions or molecules are called as ligands. Ligands donate $e^{\ominus}$s to central metal atom by forming covalent bond.

For eg:-

$[Cu(NH_3)_4]^{+2}$
↓
central metal atom

Tetramine copper (II) ion.

Transition metals can form complex compounds because their metal ions are small in size and have highe positive charge which accepts / attracts $e^{\ominus}$s from ligands. They have vacant d-orbitals to accept $e^{\ominus}$s from ligands by forming cordinate bond.

Coloured compounds Formations:-

⇒ Transition ~~element~~ metal form coloured compounds due to d-d transition. d-d transition is transition or excitation of $e^{\ominus}$s from lower energy d-orbitals to higher energy d-orbitals. It occurs due to absorption of light of certain wavelength by unpaired d-electron emitting the complementary colour.

For eg:-

light absorbe	complementary colour
Red	blue green
orange	blue
green	purple
Yellow green	violet

Ti^{+3} ⇒ colour
Ti^{+4} ⇒ colourless

show Zn^{+2} configuration
↳ No unpaired elec

Q. Why Zn^{+2} salts are colourless in nature?

Imp

Zn ⇒ [Ar] $4s^2$ $3d^{10}$

Zn^{+2} [Ar] $3d^{10}$

~~There is no unpaired electron so, it dosen't form d-d transition due to~~

Zn^{+2} ion has configuration of [Ar] $3d^{10}$. Those ions whose configuration is of $3d^0$ and $3d^{10}$ they absorp any light and Hence they are colourless. Therefore there is no d-d transition and hence no colour so, Zn^{+2} salts are colourless

Among Ti^{+3} and Ti^{+4} which one is colourless?

Here, configuration of Ti is [Ar] $3d^2 4s^2$. In Ti^{+3} 3 electrons are gone. 2 from s block & 1 from d block and configuration becomes [Ar] $3d^1$ d-d transition occurs here and is colourless but in Ti^{+4}, 2 electrons is gone from s block & 2 from d block and it's configuration [Ar] $3d^0$ There is lack of d-d transition and do not absorp any light. Therefore Among Ti^{+3} and Ti^{+4}, Ti^{+4} is colourless.

[Atomic radii :- similar atom covalenty bonded]

[used as catylist
used to make wire, medicine] use of transition element.

Atomic radii :-

Atomic radii is half of inter-nuclear distance of two covalently bonded atoms. Atomic radii decreases along the period in periodic table as no of electron increase by 1 in same shell and protons also increases (by 1), so there is high force of attraction between nucleus and valence e⁻s. Atomic radii increases along the group but it is found irregular in transition elements.

10 applications of transition elements.

1. Titanium is used in production of parts and engines of air-craft.
2. Iron is used as catalyst in Haber's process to produce ammonia.
3. Cobalt is used in paints and pigments.
4. Copper is used to make electronics wires & to make utensils.
5. Vanadium is used as catalyst (V_2O_5) in contact proc
6. Chromium & nickel are used as anti-corrosive coatings because of their corrosion-resistant properties
7. Platinum's colour of durability makes it very valuable for jewellery
8. Silver and gold are used as jewellery because of their appearence
9. Nickel is used as in producing stainless steel.

Engineering Materials & their Applications

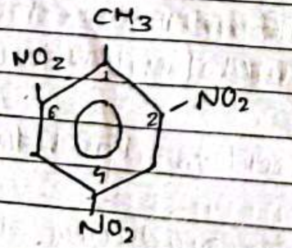
1. Explosives:- The chemical substances or mixture which when ignited undergo rapid chemical reaction releasing large amount of heat, light, sound, gases & pressure to surrounding. They are used for both destructive & constructive purposes.

They are used as weapons in war (destructive). They are used for blasting mines, rocks, making tunnels etc (constructive).

They must be cheap and safer for transportation. They must have atleast one chemical bont that can be easily broken.

i) TNT :- Trinitrotouluene :

It is a high explosive which release high amount of energy on explosion.

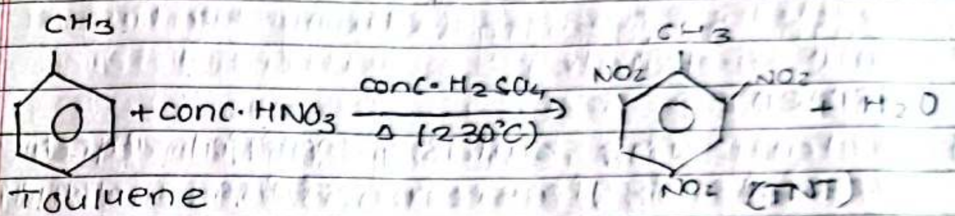


(2,4,6-trinitrotoulvene)

(hygroscopic: water [illegible] absorb gaine

Preperations:-

It is prepared by nitration of touluene at 230°C temperature. It is yellow solid. It is non-hygroscopic & is stable to metals.



It is safe for transportation so used for military purposes. It is used for excavating earth surface, construction of tunnels etc.

ii) TNG :- Trinitroglycerin

It is high explosive prepared by nitration of glycerol at 10°C

Preperation

$$\begin{matrix} CH_2-OH \\ CH_2-OH \\ CH_2-OH \end{matrix} + conc\cdot HNO_3 \xrightarrow[10^\circ C]{conc\cdot H_2SO_4} \begin{matrix} CH_2-O-NO_2 \\ CH_2-O-NO_2 \\ CH_2-O-NO_2 \end{matrix} + H_2O$$

Glycerol $\qquad\qquad\qquad\qquad$ TNG

It is heavy pale yellow oily liquid. It has low melting point (mpt) and boiling point (b.pt). It is mined with other explosives. It is used in propellants, bombs. It is used in dynamite. It is used for blasting of mines, rocks etc.

cement :-

:) The building material which consists of lime and silica and oxides of Fe, Mg, Al etc. It is finely grinded mixture of calcium silicates and aluminates. It has adhesive and cohesive property which binds sand, stones, bricks etc.

Raw materials for manufacture of cement

i. calcareous materials :- (which supply lime) lime stone, chalk etc.

ii. Argillaceous material (which supply alumina, silica) - clay, shale etc.

iii Powdered coal

iv Gypsum

composition of portland cement :-

CaO (lime 50% - 60%
SiO_2 (silica) :- 20% - 30%.
Al_2O_3 (almuina) :- 5% - 10%
Fe_2O_3 (Iron oxide) :- 3% - 5%
MgO (magnesium oxide) :- 1% to 2%
SO_3 (sulphite) : 1% to 2%
Alkali oxides (Na_2O, K_2O) :- about 1%

Manufacture of cement :-

1. Mining of raw materials :-

:) calcareous and argillaceous materials are collected and mined either by dry or wet process.

i) Dry process :- The calcareous materials are ~~eth~~ crushed and mixed with washed and dried argillaceous materials in proper ratio (moisture - 1% to 2%). This grinded and homogenized mixture is fed to rotary kiln.

rotary kiln ← [long cylindrical ~~tank~~ steel tank (fused ... to manufacture cement)

wet process:- The raw materials are mixed with water about (30% to 40%) to form slurry (paste). This slurry is fed to rotary kiln.

Pyro processing (strong heating of raw materials)

Rotary kiln is a long cylindrical steel tank which is inclined and internally lined with refratory bricks. The raw materials which is fed to kiln undergoes various rxns in different zones of kiln.

Pre heating zone:- It is upper most part of kiln where temperature is about 500°C. At this zone, moisture is removed from mixture.

Calcination or decomposition zone:- It is middle part of kiln where tempr is about 800 - 1000°C. At this zone calcium carbonate decomposes to lime.

$$CaCO_3 \xrightarrow[\Delta]{} CaO + CO_2 \uparrow$$

Dicalcium silicate	C_2S
Tricalcium silicate	C_3S
Tricalcium aluminate	C_3A
Tetra calcium ~~of~~ alumino Ferrite	C_4AF

ii) Burning (clinker formation zone) :- It is lower part of kiln where tempr is about 1400° - 1500°C. At this zone calcium aluminates and silicates are formed as below:-

$2CaO + SiO_2 \longrightarrow 2CaO \cdot SiO_2$ (Dicalcium silicate)

$3CaO + SiO_2 \longrightarrow 3CaO \cdot SiO_2$ (Tricalcium silicate)

$3CaO + Al_2O_3 \longrightarrow 3CaO \cdot Al_2O_3$ (Tricalcium aluminate)

$4CaO + Al_2O_3 + Fe_2O_3 \longrightarrow 4CaO \cdot Al_2O_3 \cdot Fe_2O_3$ (Tetra calcium alumino Ferrite (C_4AF))

This rxns takes place in cylindrical part of kiln where hot of blast air is provided by burning of powdered coal.

The small round greyish pebble like stone are obtained after burning called as clinker

3. cooling and grinding of clinker:-

=) The clinker is cooled and grinded into fine powder.

4. Addition of gypsum:-

=) Gypsum ($CaSO_4 \cdot 2H_2O$) is added to powder and homogenized. Gypsum retards quick setting of cement by forming insoluble sulphoaluminate crystals.

and

5. storing & Packing:-

=) This finely grinded mixture is stored and packed in 100 kg airtight bags.

Types of cement

1. Ordinary Portland cement (OPC):-

=) It is widely used portland cement consisting of lime, silica & gypsum. It is obtained by calcination of lime and clay. It is categorized as 33, 43, and 53 grade cement. It is used for construction.

2. ~~Pozzolong~~ Pozzolong Portland cement (PPC):-

It is obtained by mixing fly ash or volcanic ash with OPC cement. It is cheaper than that of OPC.

Setting and Hardening of cement:-

=) When cement is mixed with water it sets to hard mass. This ~~st~~ stiffening and formation of hard mass is called as setting. The gaining of strength of this hard mass is called as hardening of cement. This process takes almost 1 to 30 days. Setting and hardening of cement takes place due to hydration and crystallization reactions. The gel and crystals interlocking leads to hardness. The components C_3A, C_4AF, C_3S and C_2S undergoes setting in 1 day, 1 day, 7 days and 28 days respectively.

Exothermic rxns takes place producing gel and crystalline products. At first 24 hours tricalcium aluminate is hydrated to form gel. Gypsum retards quick setting of C_3A by forming insoluble sulphoaluminate crystals.

Also tetracalcium alumino ferrite starts to hydrate within 24 hours to form gel. $(3CaO \cdot Al_2O_3 \cdot 6H_2O)$

Then, tricalcium silicate starts to hydrate to form tobermorite gel upto 7 days. It is followed by hydration of C_2S (dicalcium silicate) forming gels and $Ca(OH)_2$ crystals upto 28 days. This leads to hardening of cement and $Ca(OH)_2$ crystals also imparts strength and maintains pH (about 12). In this way setting and hardening process 1 - 28 days

$\boxed{\text{cement min}} - \overset{1}{\boxed{C_3A \; C_4AF}} - \overset{2\text{-}7\text{ days}}{\boxed{C_3S}} - \boxed{C_2S}$ 28 days

instant setting — Final setting — Hardening

Paints :- The viscous suspension of pigment in vehicle is called paint. It is surface coating material which forms thin opaque layer on surface of wood, metal, walls etc.

charactersists of good paint:-

i) It should form uniform layer on surface

ii) It must not form voids, cracks ~~and~~ on drying.

iii) It must be resistant to moisture.

iv) It must be resistant to chemicals, heat, light, etc.

v) It must be shiny and glossy.

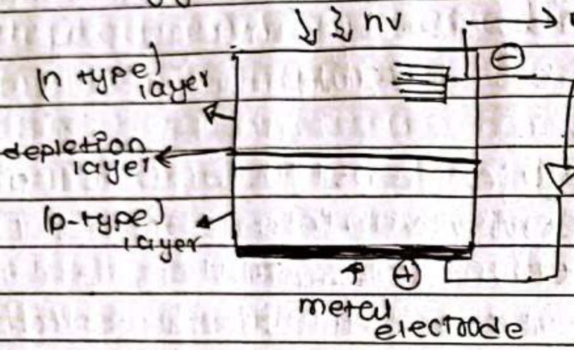
vi) It must be durable.

Various major constituents of paints are pigment, vehicle, thinner (drying oil), extenders, fillers etc.

Pigment is the solid colour forming substance of paint. It is dissolved in vehicle. For eg:- zinc oxide (white pigment) iron oxide (red pigment) vehicle consists of binder and solvent liquid which dissolves pigment. Thinner helps the paint to form the layer on surface. Extenders are the cheap materials used to increase the quantity of cement. ~~Fill~~ Fillers are oxides of metals which increase quality of paint.

Date
Page

Photovoltaic cell (solar cell):-
It is a p-n device which converts sunlight into electric current. (solar energy)
It works on the principle of photovoltaic effect. It is effect caused due to light sensitive semiconductor which converts solar energy into electric current.



construction and working of photovoltaic cell:-
Photovoltaic cell is made of two semiconductor layers (silicon). P-type layer is silicon layer doped with (Boron or Aluminium). and n-type silicon layer is doped with (nitrogen or phosphorus).

P-type layers consists of excess holes & n-type layer consists of excess electrons. The depletion layer is part of cell where two n-type and p-type layers are mixed. There is migration of e⊖s & holes. when photons (small packets of energy) strike photovoltaic cell free electrons are released. when this two ends are connected electrically there is flow of current.

Applications:-

i) Photovoltaic cells are used in storage batteries.
ii) They are used for electric supply in remote areas (as solar panels etc)
iii) They are used in watches, solar cookers etc.
iv. They are used in chargable vehicles.

Sensors:-

=) It is a device which detects the change in physical quantity like pressure, temperature, heat etc or electrical quantity like current etc.
Principle :- Different sensors work differently as they give responce to different stimulus. But basic principle includes 3-steps.

i. Detection :- It detects the change in different quantities like heat, motion, pressure etc.

ii. conversion :- It converts this physical changes into electrical signals.

iii. output :- This electric signals are recoided and processed by computer system.

POLYMERS

i) Polymers are high molecular weight compounds formed by repeated linking of small molecules (monomers). This process of combination of monomers to form polymerizn. ~~more tha~~ [more than 5 repeated units of monomers combine to form polymers]

For eg :- Polyethylene

$n(CH_2=CH_2) \xrightarrow{\text{Polymarization}} -(CH_2-CH_2)-_n$

ethene — polyethene / polythene

Types of polymers

i) Natural polymers → naturally occuring polymers For eg:- rubber, DNA etc.

ii) Synthetic polymers → artificially prepared in lab For eg:- polyethene, PVC etc.

iii) Homopolymers:- polymers formed by linking of monomers of same chemical compounds. For eg:- polyethene, PVC etc.

iv) Copolymers :- polymers formed by linking of monomers of different chemical compounds.

For eg:- Bakelite, Nylon-6,6 etc.

v) Addition polymers :- polymers formed by repeated addition of similar monomers (no biproduct)

For eg:- PVC, polyethene

vi) Condensation polymers :- formed by linking of different monomers [biproduct like H_2O, NH_3]

For eg:- Bakelite, Nylon-6,6

vii) Thermoplastic :- Polymers which can be softened on heating.

For eg:- Polyethene, PVC etc

viii) Thermosetting plastics :- Polymers which cannot be softened on heating.

For eg:- Bakelite, epoxy resins etc.

Types of polymerization

1. Addition polymerization:- The process of repeated addition of monomers one after another without formation of bi-product is called addition polymerization
=) Such polymers are addition polymers.

i. Polyethene :- It is an addition polymer formed by polymerization of ethene under high pressure (about 1500 atm) and 150-250°C temp.

$$n(CH_2=CH_2) \xrightarrow[150^\circ - 250^\circ C]{polymerization} -(CH_2-CH_2)-_n$$

ethene → polyethene

It is waxy, colourless material which can be hardened or softened according to use. It is resistant to water and chemicals.

Uses

1. It is used to make bags.
2. To prepare toys.

ii. PVC (Poly vinyl chloride) :- It is an addition polymer formed by polymerization of vinyl chloride monomer in presence of organic peroxide catalyst.

$$n(CH_2=CH) \xrightarrow[organic\ peroxide]{polymerization} -(CH_2-CH)-_n$$

vinylchloride → PVC

It is colourless and odorless. It is resistant to water, light. It is used in tank linings, water pipes, chemical carrying pipes, safety helmets, flexible bottles, cable wires etc.

3. PTFE (Teflon) (Polytetrafluoroethylene)
It is an addition polymer formed by polymerization of tetrafluoroethylene in presence of organic peroxide catalyst

$$n\left(\begin{matrix} F & & F \\ | & & | \\ C & = & C \\ | & & | \\ F & & F \end{matrix}\right) \xrightarrow[organic\ peroxide]{Polymerization} -\left(\begin{matrix} F & & F \\ | & & | \\ C & - & C \\ | & & | \\ F & & F \end{matrix}\right)-_n$$

tetrafluoroethylene → Polytetrafluoroethylene

It has high resistance, waxy in appearance. It is used in insulating wires, moto parts, cable wires, bearings, coating of non-sticky utensils etc.

condensation polymerization:-

The process of condensation of monomers to form polymers along with removal of bi-products (small molecules like H_2O, NH_3 etc) is called condensation polymen. such polymers are called condensation polymers.

1. Nylon-6,6

It is a condensation polymer formed by polymerization of adipic acid and hexamethylene diamine monomers along with elimination of water molecules.

$$n[HO-\overset{O}{\overset{\|}{C}}-(CH_2)_4-\overset{O}{\overset{\|}{C}}-OH + n[H-N-(CH_2)_6-N-H] \xrightarrow{\boxed{\text{release of } H_2O}}$$

$$-[\overset{O}{\overset{\|}{C}}-(CH_2)_4-\overset{O}{\overset{\|}{C}}-N-(CH_2)_6-\overset{H}{\overset{|}{N}}]_n-$$

It is elastic in nature } → properties
It has high strength

uses

=) It is use in making, ropes, fiber, sheets, Electric appliances

extra not in syllabus

Terylene (Dacron) polyester

It is formed by polymerization of ethylene glycol & terephthalic acid by elimin^n of water molecule.

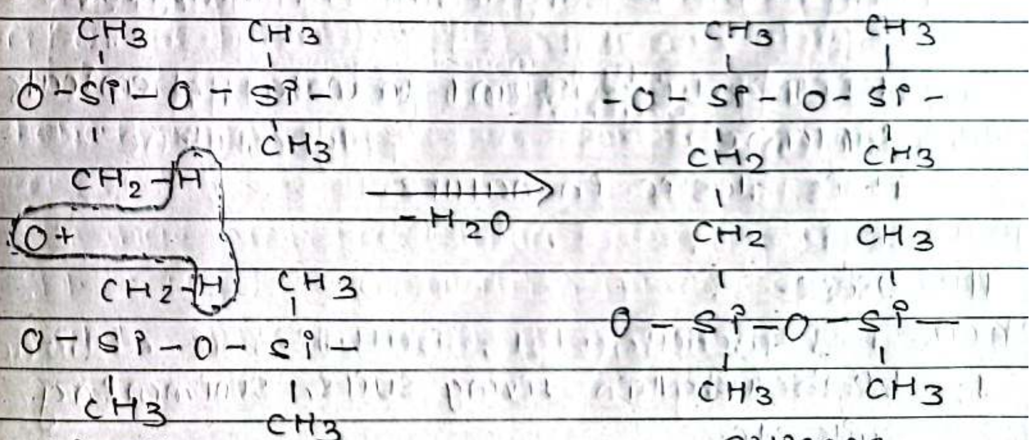
$$nC O H - CH_2-CH_2-O H + n(HO-\overset{O}{\overset{\|}{C}}-\langle O \rangle-\overset{O}{\overset{\|}{C}}-OH) \xrightarrow[-H_2O]{\text{polym}^n}$$

$$-[O-CH_2-CH_2-O-\overset{O}{\overset{\|}{C}}-\langle O \rangle-\overset{O}{\overset{\|}{C}}]_n-$$

in syllabus

Silicone rubber:-

It is formed by mixing dimethyl silicone polymers with (fillers like silic and (peroxide containing agents). The function of oxygen of peroxide is to form dimethyl bridge by elimination of H_2O



dimethyl silicone polymers

silicone rubber

It is highly resistant to heat.

uses:-

i) It is used in surgical equipments

ii) It is used in insulating parts of washing machines and electrical appliances.

iii) It is used in plastic surgery.

iv) It is used to make silicone bottles.

v) It was used in Neil Armstrong's boots (no effect due to heat variation)

Neoprene:- It is an addition polymer prepared by polymerization of chloroprene monomer.

$$n\left(CH_2=\overset{Cl}{\overset{|}{C}}-\overset{H}{\overset{|}{C}}=CH_2\right) \xrightarrow{polym^n} \left[CH_2-\overset{Cl}{\overset{|}{C}}=\overset{H}{\overset{|}{C}}-CH_2\right]_n$$

chloroprene   neoprene

It is highly resistant to water, chemicals & flame.

It is elastic in nature.

Uses:-

1. It is used in diving suits, swimming suits etc.

2. It is used in electric and cable wires.

3. It is used in gloves & PPE

Biodegradable polymers:- The polymers which are degraded by micro organism into simpler compounds water, ~~gas~~ gases, inorganic salts etc when left in environment. They can be recycled. For eg:- polylactic acid, polyglycolic acid etc.

$$n\left(HO-\underset{COOH}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}}-H\right) \xrightarrow[-nH_2O]{polym^n} \left[\underset{H}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}}-\underset{O}{\underset{\|}{C}}-O\right]_n$$

Lactic acid   Poly lactic acid (PLA)

Uses:-

i) It is used in dressing wounds.

ii) It is used in surgical equipments.

iii) It is used for food packaging materials.

iv) Used for degradable bottles, trays etc

NON- Biodegradable polymers:-

=) The polymers which cannot be degraded by microorganism when left in the environment is called non- Biodegradable polymers

For eg:- polyemone, PVC, Polyethen etc

conducting polymers:-

organic polymers which can conduct electricity are called conducting polymers. Polymers with alternate double and single bonds or presence of external ingredients like carbon black, metallonides etc cause conductivity in organic polymers.

For example:- polyacetylene, polypyrrole etc

$$\underset{\text{acetylene}}{n\,(CH \equiv CH)} \xrightarrow{\text{polymerization}} \underset{\text{polyacetylene}}{n\,[CH = CH]_n}$$

Uses:-

i) Used in rechargeable batteries as cathode ore electrolyte.

ii) used in mobile display

iii) used in sensor, photovoltaic cells etc.

Non- conducting polymers:-

=) The polymers which do not conduct electric is called non-conducting polymers. Thermoplastic and Thermosetting plastic a[nd] their bonding products are the eg of non conducting polymers.

Electrolytic cell	Galvanic cell
An electrolytic cell is a cell that uses electric current for the progression of chemical rxn.	A galvanic cell is an electro-chemical cell that can produce electricity with the help of chemical rxn.
It cannot be represented by cell notation.	It can be represented by cell notation.
There is no salt bridge.	The two electrolyte solution are connected by salt bridge.
Both electrode are dipped in same electrolytic solution.	Different electrode are dipped into different electrolytic solution.
Anode is positively charged. cathode is negatively charged.	Anode is negatively charged. cathode is positively charged.

1. What is electrochemical series and its application

=) The series of metal electrodes arranged vertically in order of their increasing standard red^n potential value is called electrochemical series. It is also activities series.

The application of electro chemical series-

1. To predict cathode and anode of galvanic cell.
2. The electrode with high reduction potential value acts as cathode while electrode with love reduction potential value acts as anode.
3. To find Emf of cell. (The difference between electrode potential of two halves cell is called Emf or cell potentials.

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

4. To find whether the cell is spontaneous or not. If the Emf of cell is positive cell is spontaneous. If the Emf of cell is negative cell is non spontaneous.

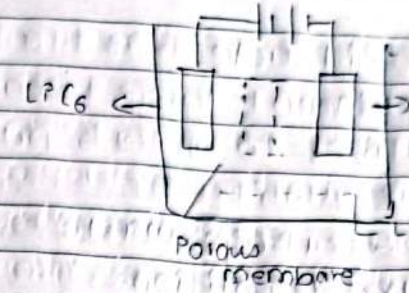
b. Explain the primary and secondary cells. Write the construction and working mechanism of lithium ion battery.

=) The cell in which spontaneous chemical rxn takes place in one direction and cannot be recharged is called primary cell. Eg:- dry cell. Dry cell is common example of primary cell. Primary cells are used in toys, calculator, watches, torchlight etc.

The cell which can be recharged again and again is called secondary cells. It is also called as rechargable battery or storage battery. It works as galvanic cell during discharging and as electrolytic cell during charging.

Lithium ion battery:

It is also called future generation battery. This second cell consist of LiC_6 (Lithium intercalated carbon electrodes) as anode and perchorate in organic solvent as electrolyte. It is widely used in laptop, smartphones etc.



Working:

At anode:

$$LiC_6 \rightarrow Li^+ + C_6 + e^-$$

At cathode

$$CoO_2 + Li^+ + e^- \rightarrow LiCoO_2$$

2
a. What is air pollution. Write its cause, effects and control measures.

=) The addition of undesirable materials into the atmosphere either by natural activities or manmade activities on the earth which adversely effect the quantity of air and disturb the dynamic equilibrium in the atmosphere is called air pollution.

The cause of ~~are~~ air polution are :-